Question Paper

Exam Date & Time: 16-Dec-2022 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH END SEMESTER EXAMINATIONS, DEC 2022

ANALOG ELECTRONICS [BME 2151]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

Instructions to Candidates:
1. Answer ALL questions.
2. Missing data may be suitably assumed.
3. Draw neat and relevant diagrams wherever necessary.

1) Making use of an NPN Silicon BJT design a switch for an input signal varying between 0V and +5V. The transistor and circuit specifications are: $ß_{dc}$ = 120, $I_{C(sat)}$ = 14mA and V_{CC} = +18V. (2)

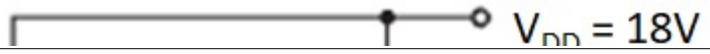
A)

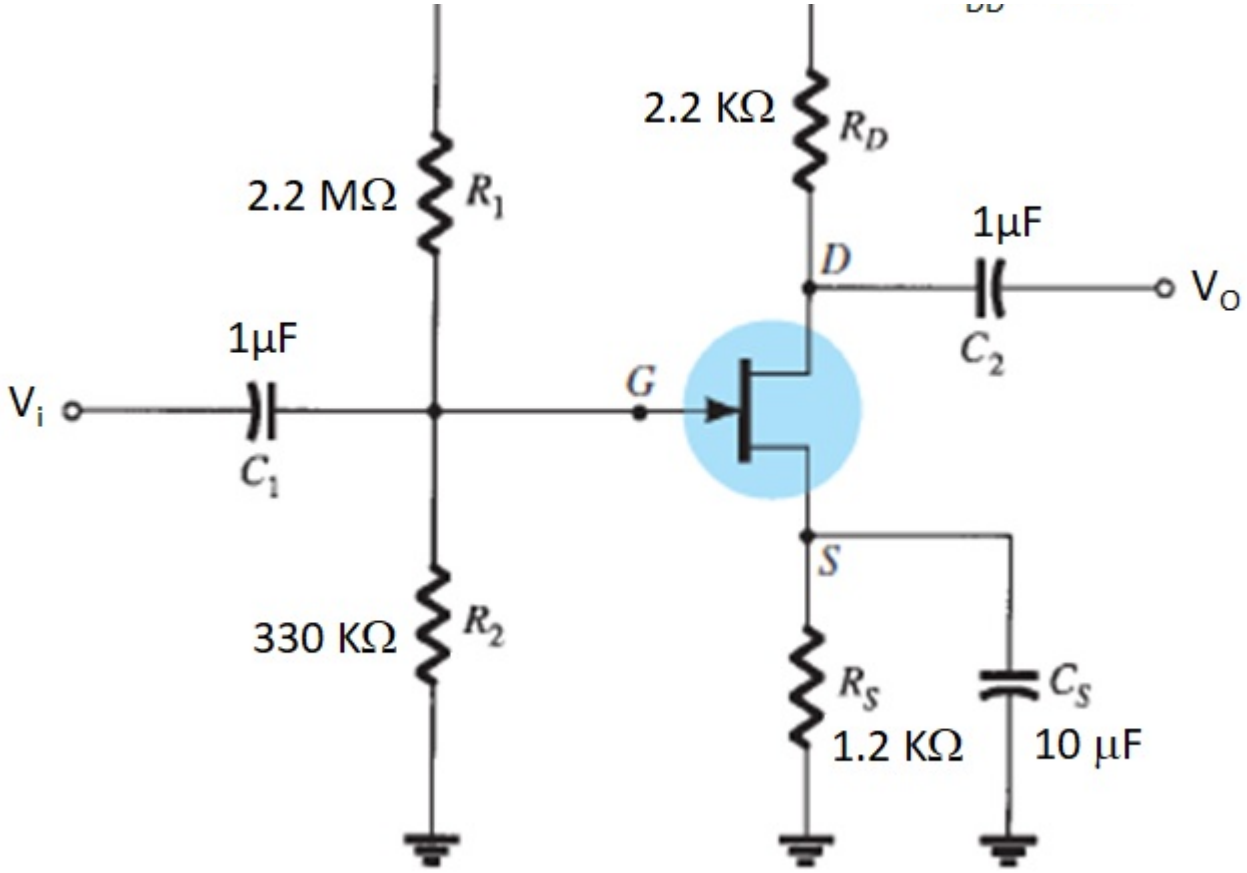
B) How do you fix the operating point of a BJT using fixed bias? Explain. What are the advantages and drawbacks of the circuit? (4)

C) How do you compensate for bias in a BJT against temperature using different techniques? Explain. (4)

2) Making use of graphical method determine I_{DQ}, V_{GSQ}, and then V_D, V_S, V_{DS} and V_{DG} of the network shown below. Specifications of the active device are: I_{DSS} = 8mA and V_P = -4V. (4)

A)





B) How do you produce saw-tooth waveform of frequency 1 KHz? Explain with neat circuit diagram and waveforms. (3)

C) Establish that the voltage gain of a BJT common emitter amplifier employing an NPN transistor in voltage divider biasing is $\frac{R_C}{r_e}$. (3)

3) Develop a small signal model of an N-channel JFET in fixed bias network, and estimate input impedance, output impedance, and voltage gain. (4)

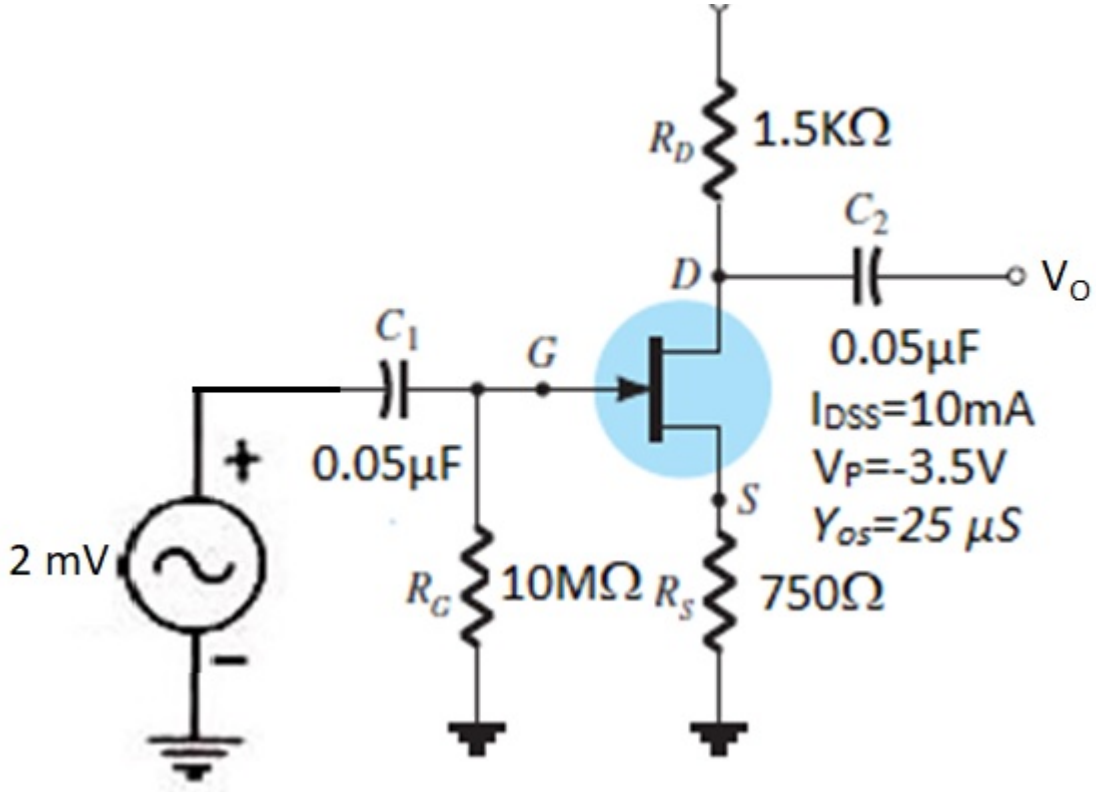
A)

B) For the JFET circuit shown below, calculate the actual output voltage with: (4)

(i) Resistor R_S bypassed.

(ii) Resistor R_S un-bypassed.

+9V



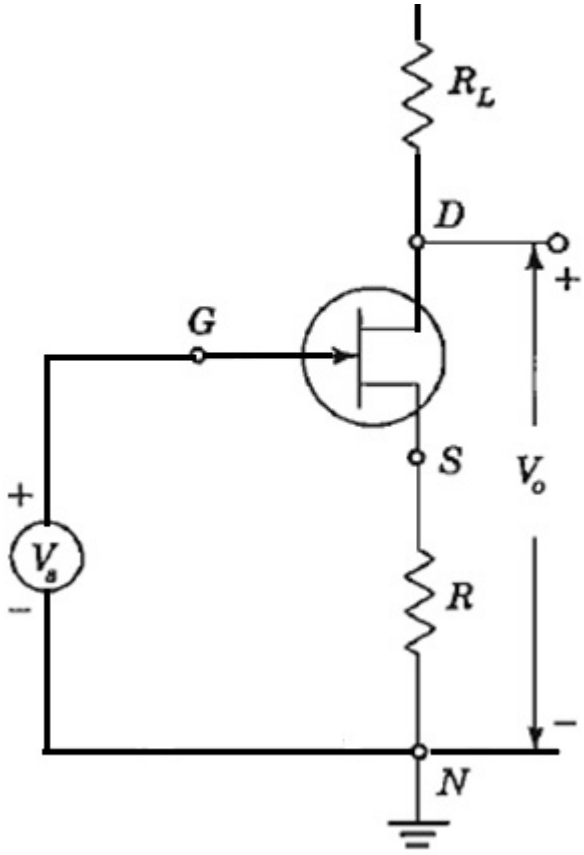
C) Compare Class-B and Class-C power amplifiers. (2)

4) With the help of block diagrams explain different types of negative feedback possible. (4)

A)

B) Analyze the following amplifier circuit for input impedance, output impedance and voltage gain with feedback. (4)

V_{DD}



C) How negative feedback modifies the characteristics of an amplifier? Explain. (2)

5) What are the conditions to be satisfied for producing sustained oscillations? Explain. (2)

A)

B) Design a JFET based RC-Phase shift oscillator for producing sustained oscillations of frequency 3KHz. Assume a capacitance of 1.0nF for the feedback network and the active device specifications are: g_m = 5mS and y_{os} = 0.25mS. (4)

C) Analyze Class-A power amplifier for: (4)

(i) A.C output power

(ii) Efficiency

And suggest a method to improve the efficiency of the amplifier.

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